Day: Date:

Section - B

— Q:2 (ii)(a) —

— A.2 (ii)(a) —

Enzymes are biological catalysts.

These metabolic reactions in living organisms are sped up by specific proteins ~~made up of~~ in the body called biological catalysts or enzymes. Enzymes are biologically active globular proteins made by living cells, which termendously speed up the biochemical reactions.

— Q.2 (iii) —

— A:2 (ii) —

Many enzymes require additional non-proteins component for their proper functioning called cofactor. Cofactor usually act as bridge between protein part of a enzyme and it's ~~substance~~ substrate or may provide energy to enzyme. There are three types of cofactors:-

Activator:-

Some enzymes require detachable inorganic ions as cofactor called activators such as Zn^{++}, Fe^{++}, Cu^{++} and chlorides ions. for example activity of salivary amylase is increased in the presence of chloride

Day:
Date:

of Chloride ions.

~~Prosth~~ Prosthetic group:-

If the cofactor is an ~~orang~~ organic molecule tightly and permenantly attached to the enzyme it is known as prosthetic group. For example, Flavin Adenine Dinucleotide (FAD), Haem group etc.

Coenzymes:-

If the cofactor is detachable organic molecule it is called coenzyme. Examples of co-enzymes are NAD (nicotinamide adenine dinucleotide), coenzyme A and vitamin A and C.

— Q:2 (iii) —

A:2 (iii)

Step 1:-

Specific substrate binds to the active site → forms Enzyme-substrate (ES) complex.

Step 2:-

Enzymes catalyses the reaction, converting substrate to protect ~~ES~~, forms EP complex (Enzyme product complex).

Day:

Date:

Step 3:-

EP complex breaks up; products ~~real~~ ~~er~~ released; enzymes is unchanged and and available again for the next reaction.

Q:2(iv)

A:2(iv)

Optimum temperature:-

Each enzyme performs best at specific temperature
: For human enzymes:- 36 → 38°c
Rate increase with temperature up to this point.

Very high temperature:-

Enzymes atom vibrate vigorously → globular structure and site damaged → denaturation (permanent loss of activity).

Very low temperature :-

Insufficient activation energy → enzymes becomes inactive (reversible - activity resumes when temperature is restored.

Day: Date:

— Q: 2 (v) —
— A: 2 (v) —

Allosteric site:-
A point on the enzyme surface other than the active site where non-competitive inhibitors attach.

Mechanism:-
Inhibitor binding alters the globular shape of the enzyme → deforms and damages the active site.

Result:-
Substrate cannot bind to the damaged active site; no ES complex forms; reaction stops.

— Q: 2 (vi) (or) —
— A: 2 (vi) (or) —

Energy storage:-
ADP + inorganic phosphate (P_i) combine by condensation to reform ATP. Energy from catabolism (or light in plants) drive this - stored in phosphate bond.

Energy release:- ATP is hydrolysed - third phosphate seperates → ~~read~~ release energy + ADP + P_i.

Day: Date:

Significance :-

Energy available on demand for movement, active transport, growth, nerve impulse, gland secretion, without ~~waset~~ ~~wat~~ wasteful overproduction.

—— Q:2(vii)(or) ——

—— A:2(vii)(or) ——

Light intensity :-

More light intensity and longer duration increase ~~photosynthetic~~ photosynthesis rate by providing more energy for light reaction

CO_2 concentration :-

CO_2 is raw material (substrate) for the photosynthetic reaction. More CO_2 increases rate, but only up to a specific limit.

Temperature :-

Optimum temperature for photosynthesis is 25°C. Temperature below or above the optimum decrease rate by affecting enzyme activity.

Day: ~~Saturday~~
Saturday

Q:2 (viii)

A:2 (viii)

Defination:-
Reactions that require light energy directly, they depends on photosystems 1 and 2 absorb light and begin photosynthesis.

Location:-
Thylakoid membrane (grana) of chloroplast, where photosynthetic ~~pigmet~~ pigments are arranged into photosystems.

Products:- ATP, NADPH and oxygen (from photolysis of water). ATP and NADPH power the calvin cycle.

Q:2 (ix)(or)

A:2 (ix)(or)

Aerobic equation:-
$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ + energy (ATP)

ATP yeild:- complete aerobic respiration yeilds 36 ATP per glucose (2 glycolysis + 2 Krebs + 32 ETC).

Comparison:- Aerobic = 36 ATP
Anaerobic = 2ATP

Requirement:- Requires oxygen as final electron acceptor; without O_2 only

Day: Date:

glycolysis + fermentation occurs.

— Q:2(x)(or) —

— A:2 (x) (or) —

Role :-

Cyclic process in mitochondrial ~~matrix~~ matrix, acetyl CoA is fully oxidised ; energy extracted as electron carriers (NADH) that feed the ETC.

Product :- $4CO_2$ + 2ATP + 6 NADH + 2 $FADH_2$ per glucose.

Significance :-

NADH and $FADH_2$ carry electrons to ETC where most ATP is produced ; CO_2 released as waste.

— Q:2(xi)(o —

— A:2 (xi) —

Alcoholic fermentation	Lactic acid fermentation
Organisms	
Yeast, some bacteria	Muscle cells ; lactic acid bacteria
Products	
Ethyl alcohol + CO_2	lactic acid only

Day: Date:

Equation

Pyruvic acid → Ethyl acid + CO_2	Pyruvic acid → lactic acid

Example

Wine making	Yougurt / cheese.

Section - C

Q:3(i)(or)

A:3 (i) (or)

Temperature:-

Rate rises with temperature to optimum (36 - 38C° for human). Above optimum: enzyme atoms vibrate - globular structure damaged → ~~det~~ denaturation (permanent). Below optimum:- Enzyme ~~is~~ inactive

PH:-

Each enzyme has specific optimum (pH). Change effects ionisation of amino acids at active site. Pepsin pH:- 2.00 (stomach). Trypsin :- pH :- 8.00 (small intenstine). Extreme pH :- denaturation.

Substrate concentration:-

Rate is directly proportional to substrate concentration ~~which~~ when enzyme molecules are sufficient. Rate increases until saturation of active sites

Day: Date:

all sites occupied. Beyond saturation, more substrate has no effect. This plateau is the ~~max~~ maximum rate (Vmax).

Inhibitors and activators :-

Inhibitors:-
(poisons, cyanide, antiboitics, drugs) reduce / stops enzyme activity.

Activators:-
Increase rate, Both regulate enzymes from inside or outside the cell.

Q.3 (ii)
A:3 (ii)

Active site :-
Charge - bearing binds to an active site → very small, specific shape, substrate binds here and reaction occurs.

Mechanisms :
Three step:-
1) Substrate binds active site → Enzyme substrate complex (ES complex)
2) Enzyme catalyses reactions → EP complex
3) Products released → enzymes unchanged.

Day: Date:

Lowering activation energy :-

Enzymes change shape/charge of substrate at active site or provides a different reaction mechanism → reactions proceed at body temperature without extreme conditions.

lock and Key model :-

Emil Fischer. Active site is rigid and fixed (Lock) ; Substrate is a specific key. Explains why only one enzyme fits only ~~ons~~ one specific substrate.

Induced fit model :-

more accepted. Active site is flexible. When the correct substrate approaches, active site changes shape to accomodate it precisely. The site is still specific, only the correct substrate can induce the correct, conformational change.

Q: 3(iii)

A: 3 (iii)

* ATP as energy currency :-

Living organism use and store energy at cellular level in the form of Adenosine triphosphate

Day: Date:

(ATP). Called Energy currency because it stores and releases energy like money -> converted b/w forms as needed. Human cells use 100 - 150 moles of ATP per day.

* Structure of ATP:-
Three components:-

1) Adenine (double ringed nitrogen base)

2) Ribose sugare (5C)
3) Three phosphate group (PO_4) in triphosphate chain, bonds b/w phosphate groups = high-energy bonds (wavy lines)
~~ATA~~ AMP = 1 phosphate
ADP = 2 phosphates
ATP = 3 phosphate.

* ATP - ADP cycle :-

Energy from light (plants

ADP + P_i - condensation

ATP hydrolysis

Energy for cellular work.

ADP + P_i

Day: Date:

* Uses of ATP :-
Working of muscles, nerve impulse transmission, growth and repair of cells, active transport. Gland secretion.

— Q.3 (iv)(or) —

Step 1 :- (glycolysis) :-
Cytoplasm. ~~Glu~~ Glucose → 2 pyruvic acid. Net yeild :- 2ATP + 2NADH. Common to aerobic and anaerobic

Step 2 :- (formation of Acetyl CoA) :-
Mitochondria. Pyruvic acid oxidised ; 2C acetyl group + ~~CO₂~~ CoA → Acetyl CoA ;

2 pyruvic acid + 2CoA → 2 Acetyl CoA + $2CO_2$ + 2 NADH.

Step 3 :- (Krebs Cycle) :-
Mitochondrial matrix. Acetyl group fully oxidised.
Yield :- $4CO_2$ + 2ATP + 6NADH + $2FADH_2$.

Step 4 :- (Electron transport Chain) (ETC) :-
Inner mitochondrial membrane. Electrons from NADH and $FADH_2$ pass through

Day: Date:

molecules releasing energy → ATP synthesis. NADH → 3 ATP each; $FADH_2$ → 2ATP ~~etc~~ each. Final electron acceptor = oxygen → combine with H_2 → water.

Glycolysis = 2ATP
Krebs cycle = 2ATP
ETC = 32ATP

36 ATP per glucose molecule.

FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

Annual Examination 2026

Biology — Class IX | Paper: SSC-I

Syllabus: Chapter 7 — Metabolism

Total Marks: 65 **Time: 2h 30m** **Version: 1**

General Instructions:

1. Write your Roll Number in the space provided on the answer sheet. Do NOT write your name.
2. Attempt ALL questions from Section A. In Sections B and C, attempt the required number choosing either the main question OR its alternative (OR).
3. All parts of a question must be answered in sequence and at one place.
4. Extra attempt of any question or any part thereof will NOT be considered.

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

#	Question	A	B	C	D	Answer
1	The sum of all chemical reactions in a living cell is called:	○ Catabolism	○ Anabolism	● Metabolism	○ Enzymology	
2	Enzymes are biologically active globular proteins produced by:	○ Dead cells	● Living cells	○ Inorganic ions	○ Ribosomes only	
3	The part of an enzyme where substrate binds and reaction takes place is the:	○ Allosteric site	○ Prosthetic group	● Active site	○ Coenzyme	
4	At very high temperature, an enzyme loses function because its structure is:	○ Activated	● Denatured	○ Phosphorylated	○ Synthesised	
5	In competitive inhibition, the inhibitor blocks enzyme activity by binding to the:	○ Allosteric site	○ Substrate	● Active site	○ Cofactor	
6	ATP is called the "energy currency" of the cell because it:	○ Is in the nucleus	● Stores and releases energy	○ Builds proteins	○ Carries oxygen	
7	ATP molecule consists of adenine, ribose sugar and:	○ Two phosphate groups	● Three phosphate groups	○ One phosphate group	○ Four phosphate groups	
8	During photosynthesis, oxygen is released as a by-product through:	○ Calvin cycle	● Photolysis of water	○ Glycolysis	○ Krebs cycle	
9	Light-independent reactions of photosynthesis (Calvin cycle) occur in the:	○ Thylakoid membrane	○ Cytoplasm	● Stroma of chloroplast	○ Mitochondrial matrix	
10	Glycolysis produces 2 pyruvic acid from glucose with a net gain of:	○ 36 ATP	○ 4 ATP	● 2 ATP	○ 38 ATP	
11	Anaerobic respiration in yeast (alcoholic fermentation) produces ethyl alcohol and:	○ Water	○ Lactic acid	● Carbon dioxide	○ Oxygen	
12	The final electron acceptor in the electron transport chain (ETC) is:	○ NADH	○ Carbon dioxide	● Oxygen	○ Water	